

DATA EXPLORATION REPORT

Document Nature

All FIR documents analyzed in this report are scanned image-based PDFs.

This means each page is not a digital text PDF but a bitmap image embedded inside a PDF container.

Implications

- Text cannot be copied directly → OCR is required
- Image quality varies due to different scanners or mobile camera captures
- Handwritten FIRs introduce natural irregularities
- Noise, skew, shadows, and stamp overlaps affect extraction
- Portrait vs landscape scans create layout detection challenges

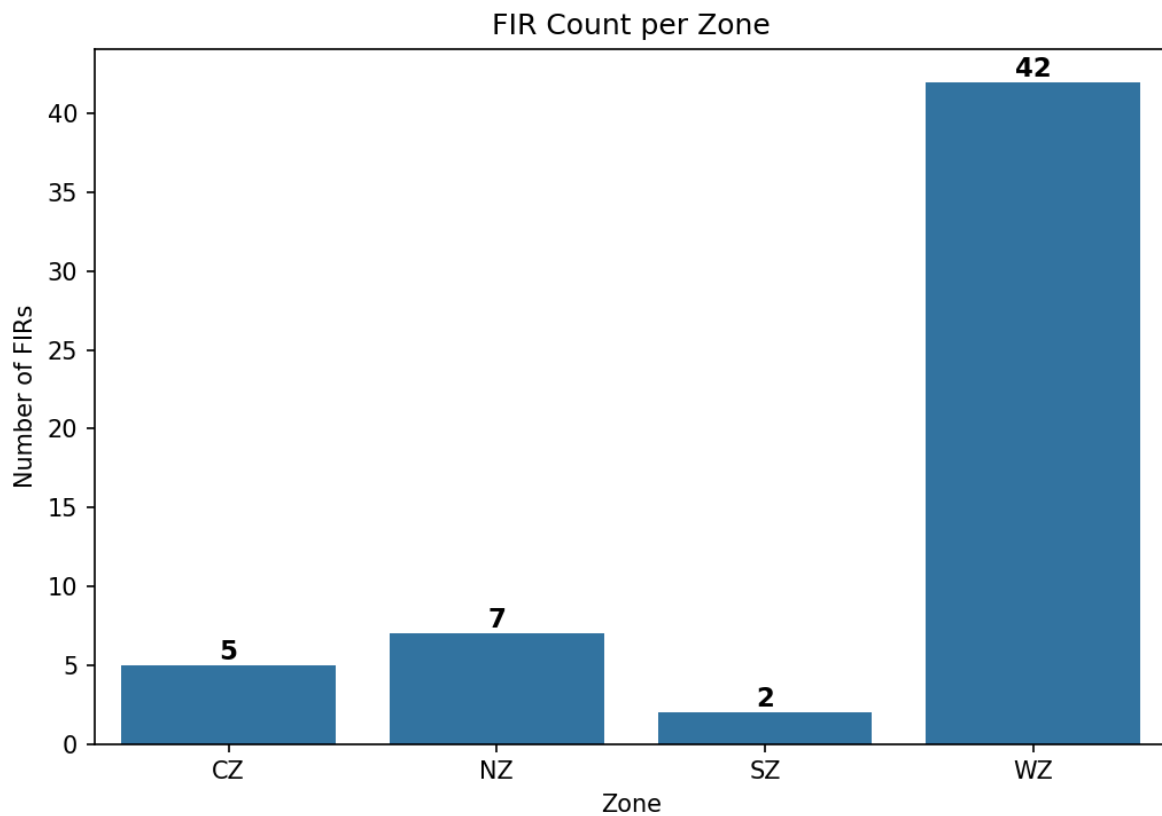
Language Information

- Most FIR text are Tamil, requiring a Tamil OCR model.
- Mixed English + Tamil also appears (dates, numbers, etc.).
- Font type and font size vary between pages.

Dataset Summary:

Total FIR PDFs : 56

This bar chart visualizes the number of FIRs collected from each zone.



From the graphs, we can conclude that:

- The **Central Zone (CZ)** has **5 FIRs**
- The **Northern Zone (NZ)** has **7 FIRs**
- The **Southern Zone (SZ)** has **2 FIRs**
- The **Western Zone (WZ)** has **42 FIRs**

Overall, **WZ clearly dominates the dataset**, contributing to **about 75%** of all FIRs, whereas **SZ has the lowest share** with just **2 FIR**.

FIR Structure Templates

1. District PS: Year: FIR No: Date:
2. Act(s) Sections
3. (a) Occurrence of Offence Day: Date From: Date To:
Time Period: Time from: Time To:
- (b) Information received at PS. Date: Time: (c) General Diary Reference:
4. Type of Information: Time:
5. Place of Occurrence: (a) Direction and Distance from PS: Beat Number:
(b) Address:
- (c) In case, outside limit of this Police Station, then the Name of P.S: District:
6. Complainant/Informant (a) Name: (c) Date/Year of Birth: (d) Nationality
(b) Father's/Husband's Name:
- (e) Passport No:
- (f) Occupation:
- (g) Address:
7. Details of Known/Suspected/Unknown accused with full particulars
8. Reasons for delay in reporting by the complainant/Informant:
9. Particulars of the properties stolen/Involved:
10. Total value of properties stolen/Involved:
11. Inquest Report/ Un-natural death Case No. If any:
12. FIR Contents:
13. Action Taken:
14. Signature/Thumb Impression of the Complainant/Informant
15. Date & Time of despatch to the court

Zone-Level with Document Type Classification

WESTERN ZONE (WZ) - 42

FIR Name	Classification
2. OCU CBCID Cr. No. 01-2015 alleged misappropriation of funds.pdf	Handwritten Document (Landscape)
3. OCU CB CID Cr. No. 02-2016 Cheating.pdf	Typed Document (Portrait)
4. OCU CBCID Cr. No. 01-2017 Criminal trespass.pdf	Typed Document (Portrait)
Cuddalore CBCID.Cr no.01_2010. Acq..pdf	Handwritten Document (Landscape)
DPI Cr.No.01-2015-FIR-Acq.pdf	Handwritten Document (Landscape)
DPI Cr.No.294-2012 FIR-MF.pdf	Handwritten Document (Landscape)
DPI Cr.No.299-2012 FIR-Acq.pdf	Handwritten Document (Landscape)
DPI Cr.No.300-2012 FIR-Acq.pdf	Handwritten Document (Landscape)
DPI Cr.No.301-2012-FIR-Acq.pdf	Handwritten Document (Landscape)
DPI Cr.No.304-2012 FIR-MF.pdf	Handwritten Document (Landscape)
DPI Cr.No.305-2012 FIR -MF.pdf	Handwritten Document (Landscape)
DPI Cr.No.306-2012 FIR -MF.pdf	Handwritten Document (Landscape)
DPI Cr.No.307-2012-FIR-MF.pdf	Handwritten Document (Landscape)
DPI Cr.No.308-2012 FIR -MF.pdf	Handwritten Document (Landscape)
DPI Cr.No.313-2012 FIR-FAD.pdf	Handwritten Document (Landscape)
Edappadi P. S Cr No. 213-2007, U.s 174Cr. P. C @ 304(A) IPC Accused was died case was charge abated on 06.08.2021.pdf	Handwritten Document (Landscape)
Erode CBCID Cr.No.01-2004 Cheating acquittal.pdf	TypeWriting (Portrait)
Erode CBCID Cr.No.01-2021 Accident case Un detected.pdf	Typed Document (Portrait)
Erode CBCID Cr.No.01-2022 Hanging “Further action dropped”.pdf	Typed Document (Portrait)
Erode CBCID Cr.No.02-2022 Hanging “Further action dropped”.pdf	Typed Document (Portrait)
Kavundapadi PS Cr.No.354-2006 Accident case Un detected.pdf	Handwritten Document (Landscape)
KGI.CBCID.Cr no.01_2012 - Acq..pdf	Handwritten Document (Landscape)
KGI.CBCID.Cr no.01_2015 Con..pdf	Handwritten Document (Landscape)
KGI.CBCID.Cr.No.01_2019 FAD.pdf	Typed Document (Portrait)
KGI.CBCID.Cr.No.01_2022 FAD.pdf	Typed Document (Portrait)
Nilgiri 2001 Dowry case.pdf	Handwritten Document (Landscape)
Nilgiri 2008 case of illegal quarrying..pdf	Handwritten Document (Landscape)
NKL CBCID CR. NO. 1_2000 - Conviction.pdf	TypeWriting (Landscape)
NKL CBCID CR. NO. 1_2009 - conviction.pdf	Handwritten Document (Landscape)
NKL CBCID CR. NO. 1_2010 - conviction.pdf	Handwritten Document (Landscape)
NKL CBCID CR. NO. 2_2015 - conviction.pdf	Handwritten Document (Landscape)
NKL CBCID CR. NO. 3_2020 Acquittal.pdf	Typed Document (Portrait)
OCU CBCID Cr. No. 01-2014 property disputed.pdf	Handwritten Document (Landscape)
Salem Cbcid Cr No. 01-2017 U.s 147, 174(a), 150(1)(a), 151 Railway act &147 IPC Acquittal on 24.02.2021.pdf	Typed Document (Portrait)
Salem Cbcid Cr No. 01-2021, U.s 420 IPC. Acquittal on 12.04.2024.pdf	Typed Document (Portrait)
Salem Cbcid Cr. No. 01-2016, U.s 406, 420, 506(II) IPC FAD on 28.06.2016.pdf	Typed Document (Portrait)
Salem cbcid Cr. No. 01-2019, U.s 174 Cr. P. C, @ 302 IPC. MF on 07.06.2021.pdf	Typed Document (Portrait)

TPR 2017 Coustody death case.pdf	Typed Document (Portrait)
TPR 2018 criminal breach of trust..pdf	Typed Document (Portrait)
TPR 2020 Cheating case.pdf	Typed Document (Portrait)
TPR 2020 Criminal trespass.pdf	Typed Document (Portrait)
TPR 2021 suicide case.pdf	Typed Document (Portrait)

SOUTHERN ZONE (SZ) - 02

FIR Name	Classification
ATC, Cr. No. 04 of 2017.pdf	Typed Document (Portrait)
CBCID ATC Cr. No. 01of 2014.pdf	Handwritten Document (Landscape)

CENTRAL ZONE - 05

FIR Name	Classification
Ariyalur CBCID Cr.No.01-23 Suspetious death.pdf	Typed Document (Portrait)
Karur Velayutham palayam PS.Cr.No.339 of 2011 Theft case UN.pdf	Handwritten Document (Landscape)
Nagapatinam Cr.No.01 of 2015 Double murder case Conviction.pdf	Typed Document (Landscape)
Thiruvarur CBCID Cr.No.01 of 2015 Women missing Conviction.pdf	Handwritten Document (Landscape)
Trichy CBCID Cr.No.02 of 2022 Suspetious drawing death.pdf	Typed Document (Portrait)

NORTHERN ZONE - 07

FIR Name	Classification
CBCID,Vellore 2022 Accidental case.pdf	Typed Document (Portrait)
Hanging case.pdf	Typed Document (Portrait)
Hqr's 2011 Forgery of DOC case.pdf	Handwritten Document (Landscape)
Metro 2018 Murder case.pdf	Typed Document (Portrait)
Metro 2022 Land Cheating case.pdf	Typed Document (Portrait)
OCU I 2022 Coustady death case.pdf	Typed Document (Portrait)
Vellore 2020 Accidental case.pdf	Typed Document (Portrait)

Page Count vs Number of FIRs:

This histogram shows how many FIRs fall into specific page lengths.



Key Findings

- **Most FIRs (24 of 56) are 2-page documents**, indicating a high level of standardization.
- Majority (**48 out of 56**) fall between **2–4 pages**.
- A few FIRs contain **10 or 11 pages**.
- Only **~5%** of FIRs exceed 6 pages.

FIRs & Its Respective Page Count:

zone	filename	page_count
CZ	Ariyalur CBCID Cr.No.01-23 Suspetious death.pdf	6
CZ	Karur Velayutham palayam PS.Cr.No.339 of 2011 Theft case UN.pdf	2
CZ	Nagapatinam Cr.No.01 of 2015 Double murder case Conviction.pdf	3
CZ	Thiruvavarur CBCID Cr.No.01 of 2015 Women missing Conviction.pdf	3
CZ	Trichy CBCID Cr.No.02 of 2022 Suspetious drawing death.pdf	4
NZ	CBCID,Vellore 2022 Accidental case.pdf	3
NZ	Hanging case.pdf	2
NZ	Hqr's 2011 Forgery of DOC case.pdf	2
NZ	Metro 2018 Murder case.pdf	3
NZ	Metro 2022 Land Cheating case.pdf	6
NZ	OCU I 2022 Coustady death case.pdf	3
NZ	Vellore 2020 Accidental case.pdf	4
SZ	ATC, Cr. No. 04 of 2017.pdf	3
SZ	CBCID ATC Cr. No. 01of 2014.pdf	2
WZ	2. OCU CBCID Cr. No. 01-2015 alleged misappropriation of funds.pdf	10
WZ	3. OCU CB CID Cr. No. 02-2016 Cheating.pdf	3
WZ	4. OCU CBCID Cr. No. 01-2017 Criminal trespass.pdf	5
WZ	Cuddalore CBCID.Cr no.01_2010. Acq..pdf	2
WZ	DPI Cr.No.01-2015-FIR-Acq.pdf	2
WZ	DPI Cr.No.294-2012 FIR-MF.pdf	2
WZ	DPI Cr.No.299-2012 FIR-Acq.pdf	2
WZ	DPI Cr.No.300-2012 FIR-Acq.pdf	2
WZ	DPI Cr.No.301-2012-FIR-Acq.pdf	2
WZ	DPI Cr.No.304-2012 FIR-MF.pdf	2
WZ	DPI Cr.No.305-2012 FIR -MF.pdf	2
WZ	DPI Cr.No.306-2012 FIR -MF.pdf	2
WZ	DPI Cr.No.307-2012-FIR-MF.pdf	2
WZ	DPI Cr.No.308-2012 FIR -MF.pdf	2
WZ	DPI Cr.No.313-2012 FIR-FAD.pdf	2
WZ	Edappadi P. S Cr No. 213-2007, U.s 174Cr. P. C @ 304(A) IPC Accused was died case was charge abated on 06.08.2021.pdf	2
WZ	Erode CBCID Cr.No.01-2004 Cheating acquittal.pdf	2
WZ	Erode CBCID Cr.No.01-2021 Accident case Un detected.pdf	3
WZ	Erode CBCID Cr.No.01-2022 Hanging "Further action dropped".pdf	3
WZ	Erode CBCID Cr.No.02-2022 Hanging "Further action dropped".pdf	3
WZ	Kavundapadi PS Cr.No.354-2006 Accident case Un detected.pdf	2
WZ	KGI.CBCID.Cr no.01_2012 - Acq..pdf	2
WZ	KGI.CBCID.Cr no.01_2015 Con..pdf	2
WZ	KGI.CBCID.Cr.No.01_2019 FAD.pdf	4
WZ	KGI.CBCID.Cr.No.01_2022 FAD.pdf	4
WZ	Nilgiri 2001 Dowry case.pdf	2
WZ	Nilgiri 2008 case of illegal quarrying..pdf	2

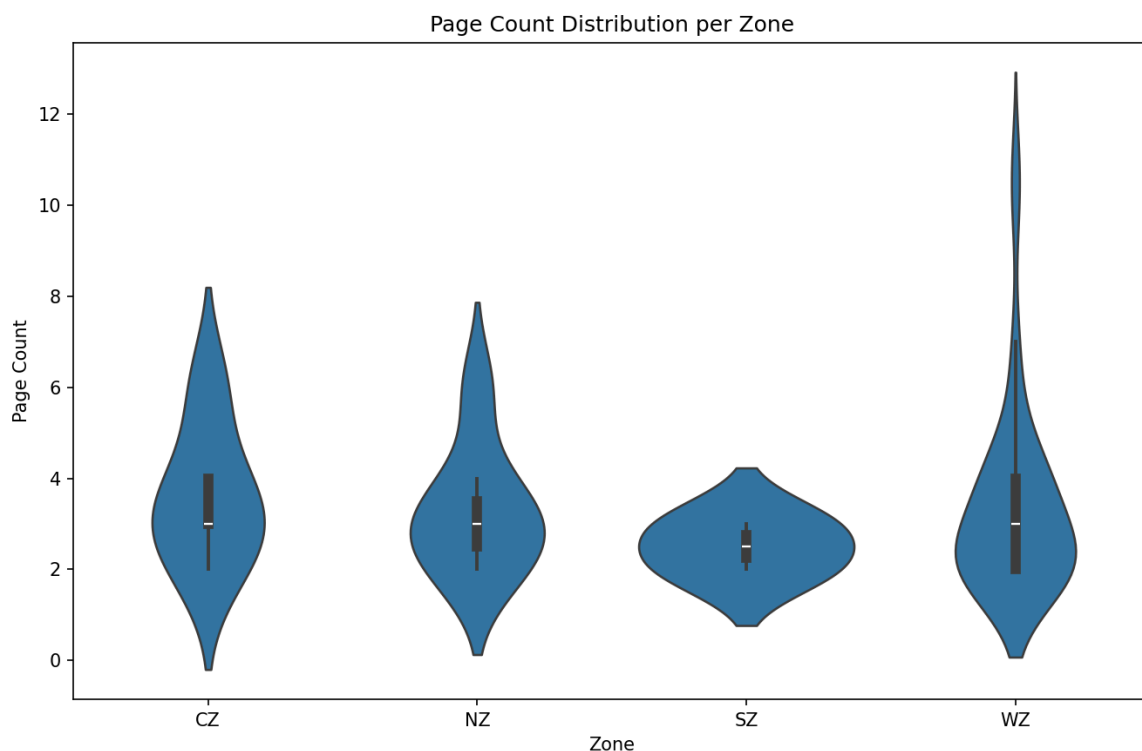
WZ	NKL CBCID CR. NO. 1_2000 - Conviction.pdf	4
WZ	NKL CBCID CR. NO. 1_2009 - conviction.pdf	4
WZ	NKL CBCID CR. NO. 1_2010 - conviction.pdf	4
WZ	NKL CBCID CR. NO. 2_2015 - conviction.pdf	2
WZ	NKL CBCID CR. NO. 3_2020 Acquittal.pdf	3
WZ	OCU CBCID Cr. No. 01-2014 property disputed.pdf	3
WZ	Salem Cbcid Cr No. 01-2017 U.s 147, 174(a), 150(1)(a), 151 Railway act &147 IPC Acquittal on 24.02.2021.pdf	4
WZ	Salem Cbcid Cr No. 01-2021, U.s 420 IPC. Acquittal on 12.04.2024.pdf	6
WZ	Salem Cbcid Cr. No. 01-2016, U.s 406, 420, 506(II) IPC FAD on 28.06.2016.pdf	5
WZ	Salem cbcid Cr. No. 01-2019, U.s 174 Cr. P. C, @ 302 IPC. MF on 07.06.2021.pdf	4
WZ	TPR 2017 Coustody death case.pdf	3
WZ	TPR 2018 criminal breach of trust..pdf	4
WZ	TPR 2020 Cheating case.pdf	11
WZ	TPR 2020 Criminal trespass.pdf	7
WZ	TPR 2021 suicide case.pdf	4

Page Distribution per Zone (Violin Plot)

What the Violin Plot Shows

This graph compares page count distribution across zones using:

- **Median (white dot)**
- **IQR / middle spread (black box)**
- **Range & outliers (thin lines)**
- **Density shape (violin form)**



Zone-Wise Observations

WZ

- Wide spread in page counts
- Contains **outliers (10+ pages)**
- Indicates varied FIR formatting styles or complexity

NZ

- More consistent distribution
- Most FIRs between **2–4 pages**

CZ

- Similar to NZ with minimal variation
- No extreme outliers detected

SZ

- Smallest sample size
- Very tight distribution
- Values clustered around **2–3 pages**